

OBJECTIVE



To optimize a method for quantifying **currently used pesticides** in blue crab tissues, using small sample amounts, to be applied in future biomonitoring studies.

RESULTS

METHOD VALIDATION



21/24 compounds optimal validation parameters [6]



Recoveries 48-200%



LOQs 0.5-10 ppb



LINEARITY >0.99

Summary of validation parameters for 21 emerging pesticides

Compounds	Haemolymph			Muscle			Haepatopancreas		
	% Recovery	% RSD	LOQ ng/mL	% Recovery	% RSD	LOQ ng/g	% Recovery	% RSD	LOQ ng/g
Propamocarb	30.4	14	0.5	54.3	9	10	41.6	10	0.5
Nitenpyram	42.4	16.2	10	66.9	6.7	5	50.8	13	5
Thiamethoxam	145	10.7	0.5	139.8	10.2	10	99.1	25.7	0.5
Clotianidin	126.9	5.9	5	142.3	6.7	5	115.8	6.5	5
Imidacloprid	135.1	14.5	0.5	143.3	14.4	0.78	120.6	10.3	0.5
Acetamiprid	138	9	0.5	166	9.6	1.03	134.7	6.4	0.5
Thiacloprid	144	8.9	0.5	168.8	7.8	0.5	114.8	5	0.5
Cymoxanil	130	12.5	5	119.5	12.5	5	113.6	10.3	5
Methyl Thiophanate	200	6.9	0.5	175	11.7	10	105.7	7.6	0.5
Spinosad-A	152.8	11.2	0.5	157.1	13.8	0.5	120.1	7.7	0.5
Spinosad-D	137.7	12.4	0.5	145.9	12	0.5	123.1	17.5	0.5
Metalaxyl	138.3	11.9	0.5	154.3	8.61	0.5	148.5	11.6	0.5
Dimethomorf	135.5	7.8	0.5	165.7	10.5	0.5	140.6	11.2	0.5
Azoxystrobin	176.7	6.4	0.5	82.5	10.5	0.5	48.3	21.3	0.5
Tebuconazole	143	7.9	0.5	138.5	10.1	0.5	120.6	9.6	0.5
Penconazole	137.7	8.6	0.5	141.7	16.7	0.5	84.8	12.4	0.5
Difenoconazole	139.7	7.5	0.5	137.5	8	0.5	118.1	9.6	0.5
Pyraclostrobin	177.4	10.6	0.5	121.8	13	0.5	150.7	15.2	0.5
Cyflufenamid	113.5	13.7	0.5	113.5	12.1	0.5	103.3	10.4	0.5
Pyriproxyfen	138.3	12.6	0.5	149.8	9.3	0.5	116.6	8.21	0.5
Pendimethalin	98.4	14.5	5	112.9	8.4	5	85.9	7.4	5

APPLICATION TO REAL SAMPLES

The method was successfully applied to samples from an **experimental** exposure to **acetamiprid** and **imidacloprid** [7]. Both insecticides were detected and quantified in all three matrices, with the highest concentrations measured in **haepatopancreas>muscle>haemolymph**. Results are shown in fig 1.

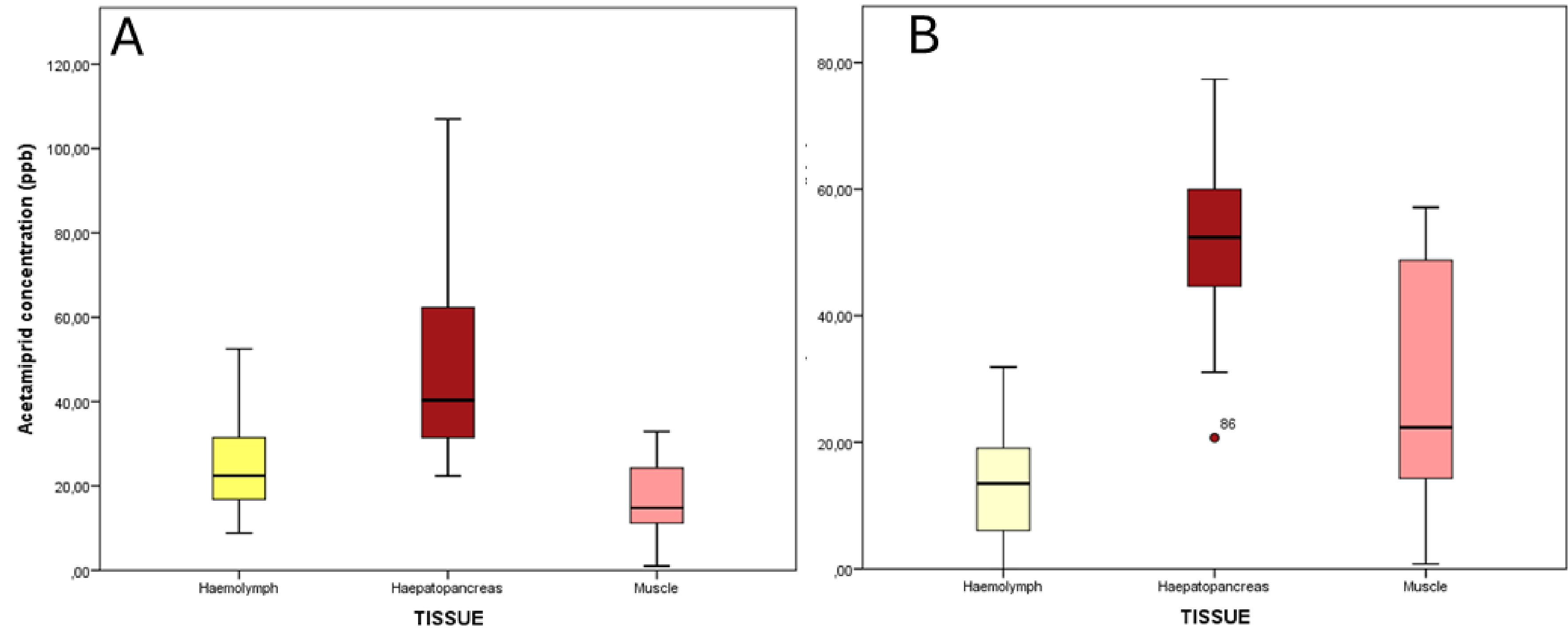


Figure 1A and 1B represent neonicotinoid concentrations found in an experimental study in the three matrices validated

INTRODUCTION



Coastal ecosystems are increasingly threatened by anthropogenic contaminants, particularly from pesticides used for **agriculture**.



Crustaceans such as the Atlantic blue crab (*Callinectes sapidus*) could be used as an alternative to bivalves to monitor contaminants, as ICES experts recommend [1].



The Mar Menor Lagoon (SE Spain) has been historically impacted by of **agriculture** and **livestock**. Nevertheless, pesticides and other agrochemicals contamination in this lagoon have been **poorly** studied [2,3].

MATERIAL AND METHODS

BLUE CRAB SAMPLING

Acclimated individuals (> 4 weeks)

SAMPLE AMOUNTS

0.2 g tissue (haepatopancreas, muscle)
0.1 mL haemolymph

EXTRACTION

Based on Ask et al [4]: 1 mL ACN:H₂O→sonication→centrifugation→filtration

ANALYSIS

HPLC-TQ [5]

METHOD VALIDATION

5 spike levels (0.5-50 ppb)
Matrix matched calibration
Triplicate analyses

APPLICATION TO REAL SAMPLES

Samples from experimental exposure to acetamiprid and imidacloprid

CONCLUSIONS

Small sample amounts required

Rapid, simple, and cost-effective extraction procedure

Sensitive and reliable for monitoring a wide range of currently used pesticides

Suitable method for future studies to monitor emerging pesticides in crab tissues.



REFERENCES

[1] Vethaak et al. 2017. Mar Environ Res, 124,:11-20.
[2] Martínez-Gómez et al. 2023. Environmental Science and Pollution Research, 30(33), 80106-80122.
[3] García-Pimentel et al. 2023. Science of the Total Environment, 905, 166828.
[4] Ask et al. 2024. Environmental Science and Pollution Research, 31(52), 61950-61958.
[5] Martínez et al. 2022. Chem Eng J 446:137182. Pesticides
[6] SANTE/12682/2019 Guidance Document on Analytical Quality Control and Method Validation Procedures for Residues Analysis in Food and Feed. 2020 European Commission Directorate General for Health and Food Safety. Safety of the Food Chain Pesticides and Biocides.
[7] Gomez-Ramirez et al.2025. In SETAC Europe 35th Annual Meeting. SETAC.